

ROOT CAUSE ANALYSIS REPORT

RCA Reference : RCA-2026-007
Related Incident: WO-4892 ? GB-14 Elevated Reading, Bay 4 (21 January 2026)
RCA Lead : Rajiv Sharma, Maintenance Supervisor
RCA Team : Kavita Desai (HSE), Deepak Singh (Maint. Engg), Anil Mehta (I&E)
Completed : 04 February 2026
Approved By : Mohan Rao, Plant Manager

1. EVENT DESCRIPTION (SUMMARY)

On 21 January 2026 at 23:15 IST, fixed gas detector GB-14 (Bay 4, Zone C) registered 22% LEL methane while Hot Work Permit HWP-2026-018 was active for plasma cutting on PV-08 in Bay 4. Operator Suresh Kumar suspended hot work at 23:15. Investigation found methane source: HX-11 Flange #7 gasket failure. Area cleared by 06:15 on 22 January.

This is the THIRD occurrence of this scenario ? elevated GB-14 readings coinciding with active hot work in Bay 4. Previous events: NMR-2019-047 (March 2019) and WO-4471 (Aug 2024).

2. TIMELINE OF EVENTS

21 Jan 2026, 21:30 ? Hot Work Permit HWP-2026-018 issued by Rajiv Sharma.
Gas reading field on permit: "0% ? area clear" (no time, no tester).

21 Jan 2026, 22:00 ? Shift change. Suresh Kumar takes over from Day Shift.
GB-14 reading at handover: 7% LEL (noted in SHL-2026-0121).

21 Jan 2026, 23:15 ? Suresh Kumar observes GB-14 at 22% LEL (rising trend over 75 min).
Immediate hot work suspension.

22 Jan 2026, 02:30 ? Anil Mehta arrives. Portable gas: 19% LEL at GB-14 location.

22 Jan 2026, 05:00 ? HX-11 Flange #7 gasket failure identified and isolated.

22 Jan 2026, 06:15 ? Area cleared (2% LEL). WO-4892 opened.

3. ROOT CAUSE ANALYSIS (5-WHY METHOD)

PROBLEM STATEMENT: GB-14 triggered at 22% LEL while hot work permit HWP-2026-018 was active in Bay 4, creating risk of ignition.

Why 1: Why was there methane at 22% LEL in Bay 4?
Answer: HX-11 Flange #7 gasket failed, allowing methane from the process stream to accumulate in Bay 4.

Why 2: Why did HX-11 Flange #7 gasket fail?
Answer: The gasket had progressive degradation from thermal cycling. Seepage was first identified in WO-4476 (August 2024) and was NOT permanently repaired ? only bolts were retorqued. No formal work order for gasket replacement was raised.

Why 3: Why was gasket replacement not completed after WO-4476?
Answer: WO-4476 was closed as "partial" with only a verbal instruction to raise a follow-up WO at "next planned shutdown." No formal WO was raised. The 2025 Annual Safety Audit (SAR-2025) flagged this as open risk but it was not escalated.

Why 4: Why was the hot work permit (HWP-2026-018) issued despite the known HX-11 seepage risk?
Answer: Rajiv Sharma was not aware that the HX-11 gasket replacement was still outstanding. There is no system linking open maintenance risks to permit issuance checks. Additionally, the gas reading on the permit was recorded without time or tester name ? it is unclear when (or whether) a fresh gas reading was taken.

Why 5: Why did the same scenario recur three times over 7 years without systemic prevention?
Answer: Each event (2019, 2024, 2026) was treated in isolation. Corrective action CA-04 from NMR-2019-047 ? a digital system to cross-link calibration status, open maintenance risks, and permit issuance ? was raised in 2019 but NEVER implemented due to budget deferral (2019, 2020, 2021, 2022) and personnel change (2023?2025). Without this system, the recurring pattern was only recognised by individuals with long institutional memory (Rajiv Sharma, Anil Mehta, Suresh Kumar).